



DPUJSONSocket

1. Static parameters:

<i>Socket_IsTCP/IP</i>	Determines if socket should communicated via TCP/IP [=true] or via UDP [=false].
<i>Socket_IP</i>	Target IP address for the sending part of socket
<i>Socket_IsServer</i>	Determines if the (TCP/IP-) socket should act as a server or as a client.
<i>Socket_PortSend</i>	Port number for the packet to be sent
<i>SendDefinition</i>	Packet definition for the packet to be sent
<i>Socket_PortReceive</i>	Port number for the packet to be received
<i>ReceiveDefinition</i>	Packet definition for the packet to be received

2. Outputs:

<i>Socket_PackageNumberSend</i>	Number of sent packets ¹
<i>Socket_PackageNumberReceive</i>	Number of received packets

3. Functional principle:

DPUsocket establishes a TCP/IP respectively UDP connection to another computer.

While the simulation is running, it exchanges one data packet with the remote computer per simulation step. A JSON protocol is used for the data packets. The content of the data packets is defined in the following form:

```
<PacketDefinition> =  
(  
    (,)(<VariableNames> , <Datatype> [,<Defaultvalue> , <Multiplier> ,  
<Offset>] )  
) ;
```

The variable names declared in the definition are provided as inputs respectively outputs of the DPUsocket. The supported data types are listed in the following table:

Data type	Data length	Min	Max
byte	1	-128	+127
ubyte	1	0	+255

¹ While in server mode, only the packets sent to the first connected socket are counted.



SILAB DRIVING SIMULATION

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word	2	-32768	+32767
uword	2	0	+65535
long	4	-2147483648	+2147483647
ulong	4	0	+4294967295
float	4	3.4e +/- 38 (7 digits)	
double	8	1.7e +/- 308 (15 digits)	